Fake News Detection with DeBERTa-v3

Engineering a Spatial Attention Approach

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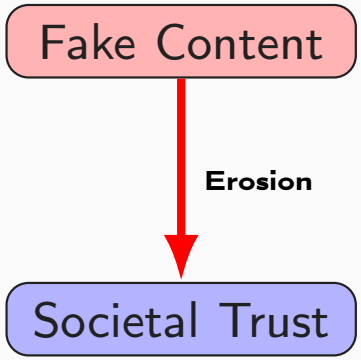
1. Motivation: Why This Project?

The Crisis of Truth

- Digital misinformation spreads 6x faster than truth.
- Fake news relies on *stylistic* cues and emotional modifiers rather than just raw facts.

Research Question

- Can we capture the **structural distance** between manipulative words better than standard models?



The Baseline Era: Bag-of-Words (BoW) & TF-IDF

The Concept: Treat text as an unordered collection of word counts.

	fake	news	mayor	denied	...
Doc 1	1	1	0	0	...
Doc 2	0	0	1	1	...
Doc 3	0	1	0	0	...

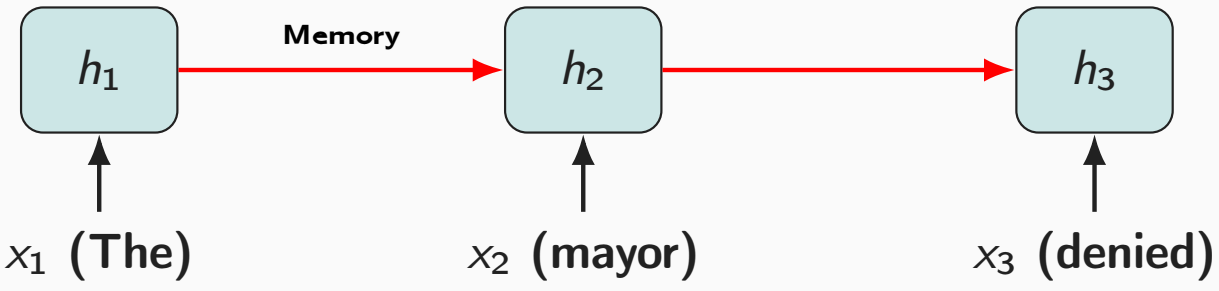
Sparse, high-dimensional matrices.

Why it fails?

- **No Context:** "Dog bites man" and "Man bites dog" have the exact same matrix.
- **No Semantics:** "Fake" and "Fabricated" are treated as completely unrelated dimensions.

The Sequential Era: RNNs & LSTMs

The Concept: Process words one by one, updating a hidden state (h_t) to carry memory forward.

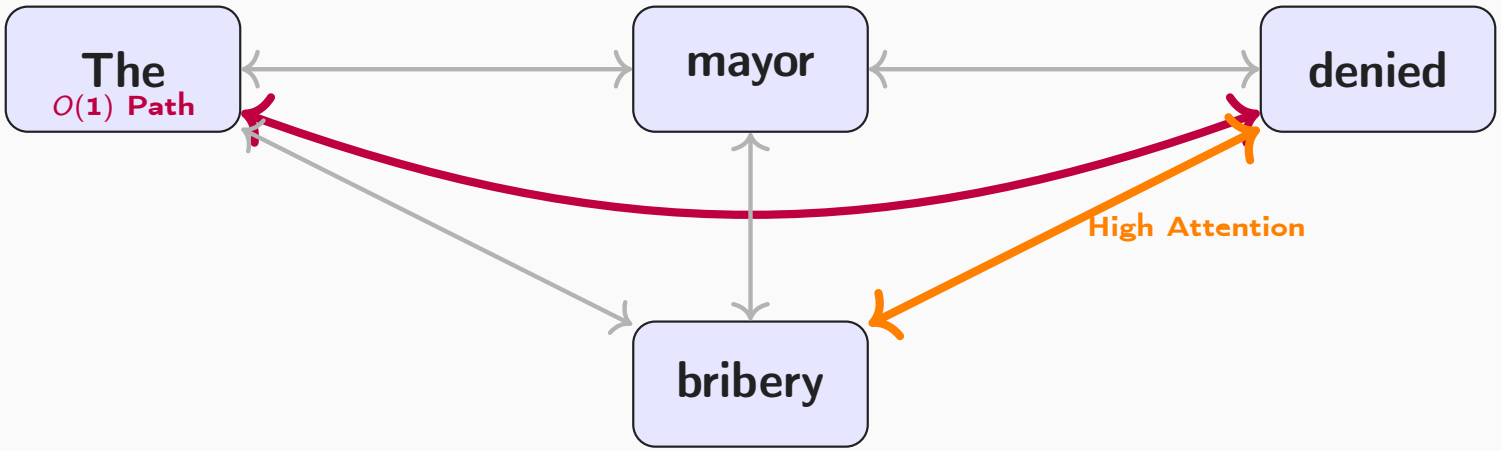


Why it fails?

- **Sequential Bottleneck:** Cannot be parallelized on modern GPUs. Training is extremely slow.
- **Vanishing Gradient:** In long fake news articles, h_{100} completely forgets what happened at x_1.

The Paradigm Shift: Transformers & Self-Attention

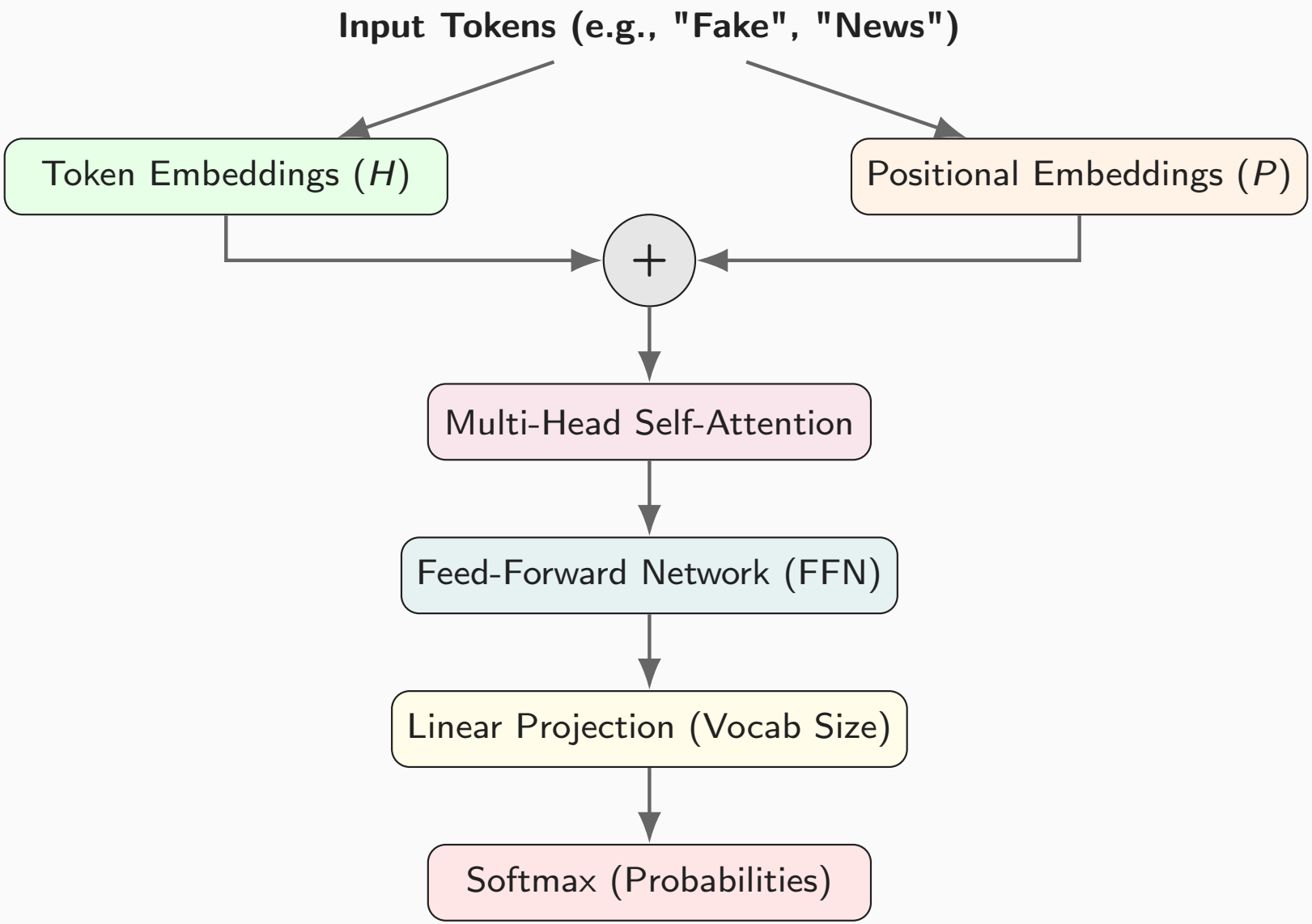
The Concept: Look at the entire context window at once. Every word asks: *"How much attention should I pay to every other word?"*



Why it wins?

- **Global Context:** The distance between word 1 and word 100 is $O(1)$ (a single mathematical jump).
- **Massive Parallelization:** No sequential waiting. Perfect for GPUs.

2. The Standard Transformer Architecture



3. The Core Concept: Knowledge vs. Position

Before diving into math, what are we actually summing in that **(+)** node?

- **Knowledge (Content / H):** What does the word mean? (e.g., "Surprisingly" implies emotion, shock).
- **Position (P):** Where is the word? (e.g., "It is the 3rd word in the sentence").

The Problem

In standard BERT, we literally add these two distinct concepts together: $X = H + P$. We force the model to analyze a "spatially-polluted meaning".

4. The Flaw of Standard Self-Attention

Let's expand the math. To get Attention Scores, we project inputs into Queries (Q) and Keys (K) using matrices W_q and W_k.

$$Q = (H + P)W_q \quad \text{and} \quad K = (H + P)W_k$$

The Attention Score is $A = Q \cdot K^T$. Let's multiply it out:

$$\begin{aligned} A &= ((H + P)W_q) \cdot ((H + P)W_k)^T \\ &= HW_qW_k^TH^T + HW_qW_k^TP^T + PW_qW_k^TH^T + PW_qW_k^TP^T \end{aligned}$$

The Fatal Flaw: Notice that W_q and W_k are used for *everything*. The model uses the **exact same weights** to understand content-to-content logic as it does for position-to-position logic!

5. "What If...?" $\rightarrow$ Disentangled Attention

What if we stopped summing H and P? What if we gave them **separate** projection weights?

Enter **DeBERTa**. It completely removes the shared $(H + P)$ entanglement.

$$
\begin{aligned}
A_{i,j} = & \underbrace{H_i \mathrm{W}_{\mathrm{q,c}} \mathrm{W}_{\mathrm{k,c}}^{\mathsf{T}} H_j^T}_{\text{Content-to-Content}} \\
+ & \underbrace{H_i \mathrm{W}_{\mathrm{q,c}} \mathrm{W}_{\mathrm{k,r}}^{\mathsf{T}} P_{i|j}^T}_{\text{Content-to-Relative Pos}} \\
+ & \underbrace{P_{j|i} \mathrm{W}_{\mathrm{q,r}} \mathrm{W}_{\mathrm{k,c}}^{\mathsf{T}} H_j^T}_{\text{Relative Pos-to-Content}}
\end{aligned}
$$

Now, the model has specialized "experts" (W_c for content, W_r for space) to calculate attention precisely!

6. A Concrete Numerical Example

Let's prove the difference with a tiny 1x2 matrix example for two tokens: a Query (token q) and a Key (token k).

The Data:

- Content: $H_q = [1, 0]$ *(meaning A)*, $H_k = [0, 1]$ *(meaning B)*.
- Position: $P_q = [1, 1]$, $P_k = [1, 1]$ *(they share the same generic spatial zone)*.

The Goal: We want the model to output an attention score of **0** because their meanings ($[1, 0]$ and $[0, 1]$) are completely unrelated (orthogonal).

7. Standard vs. Disentangled (The Showdown)

Standard Attention: (Assuming $W = I$ for simplicity)
$X_q = H_q + P_q = [1, 0] + [1, 1] = [2, 1]$
$X_k = H_k + P_k = [0, 1] + [1, 1] = [1, 2]$
Score $= X_q \cdot X_k^T = (2 \times 1) + (1 \times 2) = \mathbf{4}$.
(Fail! It paid attention just because positions were similar!)

Disentangled Attention (DeBERTa):
Model learns distinct weights. It sets $W_c = I$, but learns $W_r = 0$ (ignoring position for this specific task).

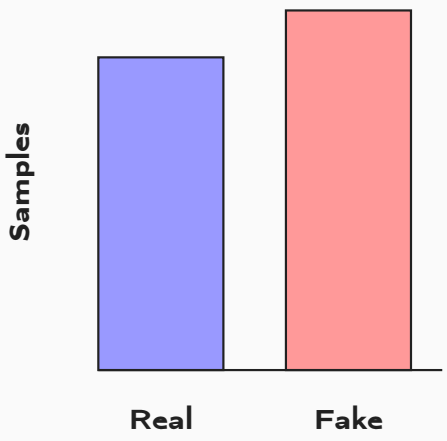
$$\begin{aligned}\text{Score} &= (H_q I H_k^T) + (H_q I \cdot 0 \cdot P_{q|k}^T) + \dots \\ &= ([1, 0] \cdot [0, 1]^T) + 0 + 0 \\ &= \mathbf{0}.\end{aligned}$$

(Success! The model successfully isolated meaning from spatial noise.)

8. Dataset & Experimental Setup

ISOT Fake News Dataset (Ahmed et al.)

- **Balance:** Near 50/50 split (45k total).
- **Real:** Reuters (Factual, precise).
- **Fake:** Clickbait (Sensational).
- **Constraint:** Max length 256 tokens. Focused on the "Inverted Pyramid" structure of journalism.

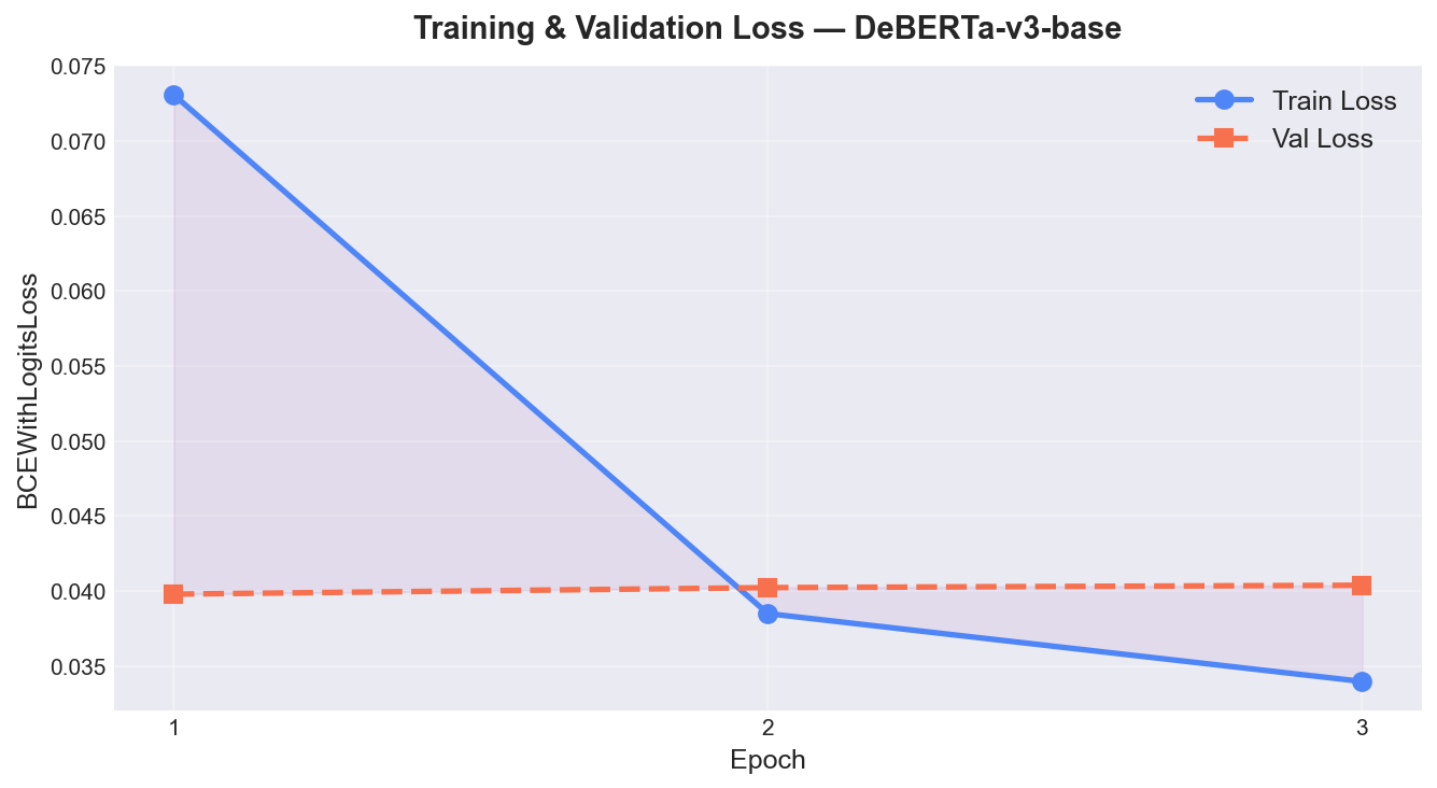


9. Engineering: 184M Parameters vs. 6GB VRAM

The Hardware Challenge
Standard fine-tuning of DeBERTa-v3 on consumer GPUs yields immediate Out-Of-Memory (OOM) crashes.

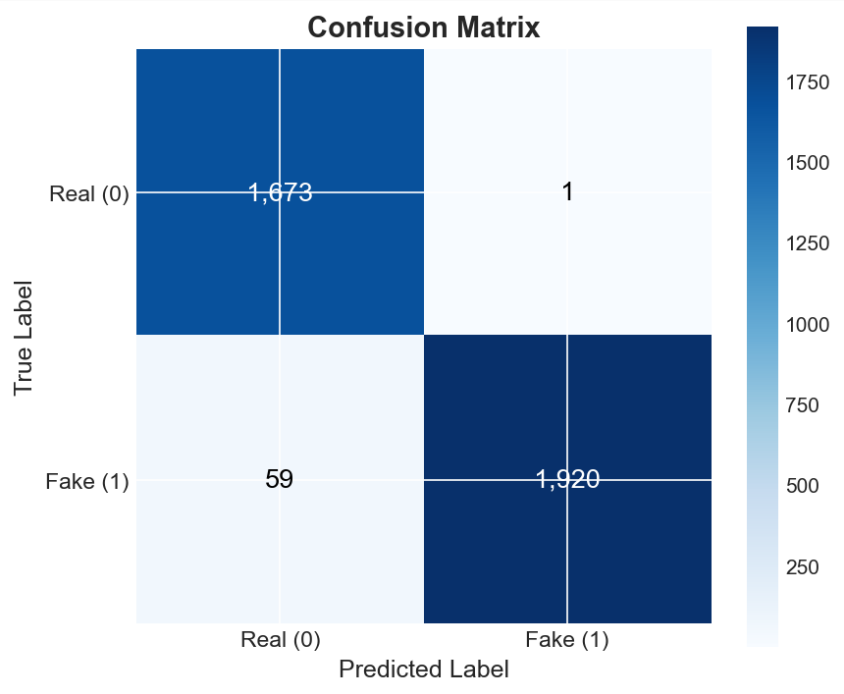
1. **FP16 Mixed Precision (`torch.amp`):** Reduces tensor memory footprint by 50% while preserving FP32 gradients.
2. **Gradient Accumulation:**
   - Physical Batch Capacity: 4
   - Accumulation Steps: 4
   - **Effective Batch Size: 16** (Ensures stable descent).
3. **Classification Head:** Replaced multi-class Softmax with a mathematically stable single-neuron `BCEWithLogitsLoss`.

10. Results: Learning Convergence



Observation: Validation loss follows training loss perfectly. The model learned generalizable linguistic rules, not dataset-specific noise.

11. Results: Confusion Matrix & Metrics



Metric	Score
Accuracy	98.36%
F1 Score	0.9846
Precision	99.95%
Recall	97.02%
ROC-AUC	0.9989

- **Key Insight:** Near-zero False Positives (1). Crucial for avoiding the censorship of legitimate news.

12. Qualitative Verification (Manual Inference)

Case 1: Reuters

"WASHINGTON (Reuters) - The U.S. Senate voted on Thursday to approve the new defense budget, aiming to restructure current military expenditures following weeks of intense bipartisan negotiations."

Result: REAL (1)
Logit: **11.07**
Confidence: 100.0%

Case 2: Clickbait

"BREAKING: You will not believe what they just found! Shocking new leaked documents completely destroy the official narrative. The mainstream media is trying to cover this up immediately!"

Result: FAKE (0)
Logit: **-9.59**
Confidence: 99.99%

Conclusion: Disentangled Attention successfully identified the spatial proximity of emotional adjectives to factual nouns as a "Fake" heuristic.

13. Conclusion & Future Work

- **Achievement:** Deployed state-of-the-art NLP classification on highly constrained hardware via rigorous optimization.
- **Limitation:** Operates purely on linguistic heuristics. It cannot verify *actual world events*.
- **Future Work (RAG):** Use this model as a high-speed primary filter. Connect flagged articles to a **Retrieval-Augmented Generation** system with internet access for definitive fact-checking.

Thank you!

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